

HUANXUAN LIAO

✉ huanxuanliao@gmail.com · ☎ 15575986157 · 🌐 [Xnhyacinth](https://xnhyacinth.github.io/) · 🔗 <https://xnhyacinth.github.io/>

🎓 EDUCATION

Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing 2023 – Present

PhD student in Pattern Recognition and Intelligent Systems (NLPR). Advisors: [Shizhu He](#) and [Kang Liu](#)

North China Electric Power University (NCEPU), Beijing

2019 – 2023

Bachelor of Engineering in Intelligence Science and Technology (Innovation Class). Ranked 1/13

💡 RESEARCH INTERESTS

I work toward **foundation models for long-horizon adaptive intelligence**: models that use growing contexts effectively, retain reusable capabilities, and act reliably over extended interactions.

- **Long-Context Data & Model Co-Design**: long-context data curation and synthesis, length- and capability-aware mixtures, scalable pretraining and mid-training, efficient attention, KV-cache optimization, and architectures to jointly improve long-context capability, scalability, and efficiency ([LCLM-Horizon](#), [HyCo₂](#), [SparK](#), [ResAdapt](#), [AllocEmbed](#)).
- **Knowledge & Continual Adaptation**: acquiring, transferring, internalizing, and updating knowledge and capabilities across evolving tasks, while balancing plasticity, retention, and generalization ([TAGI](#), [AAG](#), [NesyCD](#), [SKIntern](#), [SpARTA](#), [SHIP](#), [DyPRAG](#)).
- **Agent Learning & Self-Improvement**: persistent memory, long-horizon decision-making, and learning from interaction and feedback to verifiably refine agent memory, policies, and learning procedures ([CITI](#), [KATE](#), [OPERATE](#), [Distill Yourself](#), [SLE](#), [BuPO](#), [Tool-RL-Box](#)).

👥 RESEARCH EXPERIENCE

Tencent (Project UP) Beijing

Jul. 2026 – Present

Pretraining Data and Models

- **Long-context pretraining data**: Curated long-context pretraining corpora: **186.8B tokens** in six 8K–512K bins plus a separate **10B-token** control set; validated data coverage, answer positions, loss masks, and length distributions.
- **Data recipe and training analysis**: Compared ten recipes under matched training across 8K–128K. In a 10B-model ablation, answer-boundary supervision with a final query raised peak aggregate MRCR from **0.854 to 0.911**.
- **Sparse architecture exploration**: Evaluated long-context quality and prefill/decode cost at matched models and budgets; lower selector FLOPs did not ensure end-to-end speedup because selection overhead and KV movement remained.
- **Vision-code pretraining data**: Contributed image-code checks, cross-library perceptual deduplication, and Qwen3-VL-Embedding / SVGFaith-Score screening for SVG, sketch-to-HTML, and CAD pretraining data; SVG contained **8.13M samples** (about **41B tokens**).

ByteDance Beijing

Jun. 2025 – Jun. 2026

Multimodal Understanding

- **Production AIGC detection**: Deployed multimodal representations for TikTok AIGC detection: online F1 improved by **>3%** and compute fell by **30%–50%**; reused the representations for embedding retrieval.
- **Resolution before encoding**: Proposed [ResAdapt](#) and CAPO for frame-wise budgets before visual encoding: **>90%** fewer visual tokens, up to **16×** more frames at matched compute, and **>15%** relative long-video reasoning gains.
- **Video embedding input allocation**: Contributed to [AllocEmbed](#), a lightweight preview-based frame-resolution allocator trained with retrieval-driven policy optimization (RDPO) before a frozen embedding backbone. At matched visual budgets, LongRet mean Hit@1 rose by **5.69–9.09 percentage points** across four backbones (mean over four subsets), transferring without retraining.
- **Compression and architecture evaluation**: Compared visual-token and KV compression for accuracy–compute tradeoffs; explored skip-layer and dynamic-boundary designs.
- **Long-horizon agent evaluation**: Built [OPERATE](#) to evaluate observation, intervention, and plan persistence in evolving simulators, using evidence-based scoring and same-seed counterfactual checks.

- **Hybrid context compression:** Proposed [HyCo₂](#) to retain global semantics and local evidence; matched uncompressed performance with about **88.8%** fewer input tokens and averaged a **13.1% relative gain** across seven knowledge-intensive QA tasks.
- **Long-context research:** Core contributor to the [LCLM-Horizon survey](#) (data, architectures, training, evaluation); **Curator & Maintainer** of [Awesome-LCLM](#).

SELECTED PUBLICATIONS [FULL LIST]

Long-Context Data & Model Co-Design

[ACM TOIS'26] [Huanxuan Liao](#), Yao Xu, Shizhu He, Jun Zhao, Kang Liu. [HyCo₂: Hybrid Context Compression with Global Semantics and Local Evidence](#).

[arXiv'26] [Huanxuan Liao](#), Zhongtao Jiang, Yupu Hao, Yuqiao Tan, Shizhu He, Ben Wang, Jun Zhao, Kun Xu, Kang Liu. [ResAdapt: Adaptive Resolution for Efficient Multimodal Reasoning](#).

[arXiv'26] Song Jin, Zhongtao Jiang, Chenglei Shen, [Huanxuan Liao](#), Haozhe Chi, Zhiwei Wang, Kun Xu, Yong Liu. [Allocate Before You Embed: Adaptive Visual Input Allocation for Video Embeddings](#).

[AAAI'26] [Huanxuan Liao](#), Yixing Xu, Shizhu He, Guanchen Li, Xuanwu Yin, Dong Li, Emad Barsoum, Jun Zhao, Kang Liu. [Spark: Query-Aware Unstructured Sparsity with Recoverable KV Cache Channel Pruning](#).

[arXiv'25] Jiaheng Liu, Dawei Zhu, Zhiqi Bai, Yancheng He, [Huanxuan Liao](#), Haoran Que, Zekun Wang, et al. [A Comprehensive Survey on Long Context Language Modeling](#). (Core Contributor)

Knowledge & Continual Adaptation

[ACL'26] [Huanxuan Liao](#), Shizhu He, Yupu Hao, Yequan Wang, Wenhao Teng, Xiangwen Liao, Jun Zhao, Kang Liu. [Spectral Disentanglement: Rank-Aware Task Adaptation for Rehearsal-free Continual Learning in LLMs](#).

[ACL'26] Wangtao Sun, Haotian Xu, [Huanxuan Liao](#), Xuanqing Yu, Zhongtao Jiang, Shizhu He, Jun Zhao, Kang Liu. [Shuttle Between Symbolic Instructions and Neural Parameters of Large Language Models](#).

[NeurIPS'26] Yuqiao Tan, Shizhu He, [Huanxuan Liao](#), Jun Zhao, Kang Liu. [Dynamic Parametric Retrieval Augmented Generation for Test-time Knowledge Enhancement](#). *TTCL Workshop*.

[AAAI'25] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, Jun Zhao. [Neural-Symbolic Collaborative Distillation: Advancing Small Language Models for Complex Reasoning Tasks](#).

[COLING'25] [Huanxuan Liao](#), Shizhu He, Yupu Hao, Xiang Li, Yuanzhe Zhang, Jun Zhao, Kang Liu. [SKIntern: Internalizing Symbolic Knowledge for Distilling Better CoT Capabilities into Small Language Models](#).

[COLING'25] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Shengping Liu, Kang Liu, Jun Zhao. [Awakening Augmented Generation: Learning to Awaken Internal Knowledge of Large Language Models for Question Answering](#).

[NeurIPS'24] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Yanchao Hao, Shengping Liu, Kang Liu, Jun Zhao. [From Instance Training to Instruction Learning: Task Adapters Generation from Instructions](#).

Agent Learning & Self-Improvement

[EMNLP'26] Yuqiao Tan, Minzheng Wang, Shizhu He, [Huanxuan Liao](#), Chengfeng Zhao, Qiunan Lu, Tian Liang, Jun Zhao, Kang Liu. [Bottom-up Policy Optimization: Your Language Model Policy Secretly Contains Internal Policies](#). **Oral**.

[ICML'26] Yupu Hao, Zhuoran Jin, [Huanxuan Liao](#), Kang Liu, Jun Zhao. [Why Multi-Step Tool-Use Reinforcement Learning Collapses and How Supervisory Signals Fix It](#). *FAGEN Workshop*.

♥ AWARDS AND SCHOLARSHIPS

National Scholarship ×3

2021 / 2022 / 2025

Beijing Outstanding Graduate

2023.06

🔗 PROFESSIONAL SERVICES

Conference Reviewing: ACL ARR 2024–2026. NeurIPS 2025–2026. ICML 2026. COLM 2026. ICLR 2025–2026. AAAI 2026.

Open Source: [Awesome-LCLM](#): Curator & Maintainer; [NVIDIA/kvpress](#): Contributor; [Distill Yourself](#): Core Logic & Code Contributor.